

ABDUL SALAM

AI And Backend System Engineer | AI Associate | Backend Engineer

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Summary

Computer Science graduate specializing in AI engineering and backend development with hands-on experience building AI-powered applications, production APIs, machine learning systems, and RAG pipelines. Experienced with Python, FastAPI, Django, REST APIs, LLM integration, RAG, vector databases, embeddings, machine learning, and computer vision. Built and deployed end-to-end systems involving PostgreSQL, Redis, Celery, Docker, Supabase, Streamlit, and cloud platforms. Currently expanding practical expertise across backend AI engineering, machine learning, data analysis, model serving, and production-oriented AI systems.

Technical Skills

Programming: Python, SQL, JavaScript

Backend & APIs: FastAPI, Django, Django REST Framework, REST APIs, Pydantic, SQLAlchemy, WebSockets, Django Channels, JWT, Role-Based Access Control

AI & LLM Engineering: LLM Integration, RAG, Prompt Engineering, Groq API, Gemini API, Sentence Transformers, Hugging Face, ChromaDB, Embeddings, Semantic Search, AI Fluency

Machine Learning & Data: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, EDA, Data Cleaning, Feature Engineering, Regression, Classification, Clustering, Model Evaluation

Computer Vision: PyTorch, YOLOv8, OpenCV, Deep SORT

Databases & Infrastructure: PostgreSQL, SQLite, ChromaDB, Redis, Neon PostgreSQL, Supabase

DevOps & Tools: Docker, Docker Compose, Celery, Git, GitHub, Postman, Swagger/OpenAPI, Render, Railway, Streamlit, Vercel, Gunicorn, Daphne

Frontend: React, Vite, Streamlit, Tailwind CSS, HTML, CSS

Experience

Backend AI Engineer Intern

Jul 2026 – Present – GitHub

FlyRank AI – Remote

- Developed backend services using Python and FastAPI, progressing from REST API fundamentals to production-oriented backend and AI engineering patterns.
- Implemented persistence architecture using PostgreSQL and repository abstractions, allowing storage implementations to be changed without modifying route or service logic.
- Built asynchronous AI processing with Celery and Redis, moving long-running Groq LLM calls into background workers with job-status tracking.
- Implemented idempotent job submission, retry logic with exponential backoff, and failure alerting for reliable AI background processing.
- Integrated Supabase authentication with FastAPI, implementing JWT verification, protected routes, reusable authentication dependencies, and role-based authorization.
- Worked with LLM APIs, prompt engineering, RAG concepts, vector stores, and embeddings, with an emphasis on provider-agnostic AI integration.
- Containerized multi-service backend systems using Docker and Docker Compose and documented setup, testing, and verification workflows.
- Applied environment-based configuration, deployment practices, API documentation, and verification-first development across backend assignments.

Associate AI Engineer – AI Bootcamp

Jul 2026 – Present – Sprint 01 | Sprint 02

Nolyth Software House – Remote

- Built backend applications using Python, FastAPI, SQLAlchemy, PostgreSQL, JWT authentication, and Streamlit, progressing from backend foundations into applied machine learning.
- Developed and deployed a full-stack Task/To-Do Manager with authenticated FastAPI APIs, database persistence, Streamlit UI, and connected cloud services.
- Practiced data analysis and visualization using NumPy, Pandas, Matplotlib, and Seaborn, including data cleaning, exploratory analysis, feature creation, and statistical interpretation.
- Implemented and evaluated machine learning models including Logistic Regression, Decision Tree, Random Forest, KNN, and SVM across practical datasets.
- Built a deployed Titanic Survival Prediction application serving trained ML models through FastAPI with PostgreSQL persistence and a Streamlit interface.
- Worked through regression, classification, clustering, model evaluation, and feature-engineering workflows as part

of the AI engineering curriculum.

- Applied production-oriented practices including environment isolation, API testing through Swagger, deployment debugging, database connection handling, and documented technical workflows.

Django Backend Developer Intern

Nov 2025 – Jan 2026

Developers Hub Corporation – Remote

- Built and maintained backend functionality for a production-oriented Django eCommerce application, including models, views, REST APIs, and product/order workflows.
- Implemented JWT authentication, PostgreSQL data modelling, and user-based access control for secure backend operations.
- Worked collaboratively in a remote engineering environment, contributing to debugging, feature implementation, and backend architecture decisions.

Selected AI & Software Projects

Job Connect – AI-Powered Job Portal

Final Year Project

Django, DRF, Channels, WebSockets, Redis, ChromaDB, Sentence Transformers, Groq, PostgreSQL

- Built and deployed a full-stack job marketplace supporting Job Seekers, Employers, and Admins with role-based authentication, job management, applications, resume management, and real-time messaging.
- Developed an AI Resume Intelligence system covering resume parsing, ATS scoring, career analysis, semantic job matching, and AI-generated personalized cover letters.
- Implemented semantic job recommendations using Sentence Transformers and ChromaDB, combined with structured scoring to rank relevant opportunities.
- Integrated Groq LLM capabilities for AI-generated career and resume-related functionality.
- Resolved production deployment challenges involving Redis, PostgreSQL, ChromaDB persistence, and cloud deployment.

DocuMind – AI Legal & Financial Document Analyst

Django, RAG, ChromaDB, Sentence Transformers, Groq LLM, pypdf, PostgreSQL

- Built an end-to-end RAG pipeline for uploading, extracting, chunking, embedding, storing, and semantically retrieving information from user documents.
- Integrated Sentence Transformers and ChromaDB for embedding generation and vector-based document retrieval.
- Integrated Groq LLM responses with grounded prompts to answer questions within the context of uploaded documents and provide source references.
- Implemented persistent, user-scoped document chat history with environment-driven database configuration and PostgreSQL production readiness.

VisionX Tracker – Real-Time Object Detection & Tracking

YOLOv8, PyTorch, OpenCV, Deep SORT, Streamlit

- Built a real-time computer vision system using YOLOv8 and Deep SORT to detect and track 10+ object classes with persistent object IDs.
- Implemented multiple input modes including webcam, video files, and static images with a real-time analytics interface.
- Deployed the application using Streamlit Cloud for public demonstration.

Education

BS Computer Science

Feb 2022 – Feb 2026

COMSATS University Islamabad – Vehari Campus

Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Digital Image Processing, Data Structures, Object-Oriented Programming, Web Development, Software Engineering.

Certifications

Building REST APIs with Django • Deloitte Australia Technology Simulation – Forage • Asynchronous Python for Real-World Apps – LinkedIn Learning • Designing RESTful APIs – LinkedIn Learning

Soft Skills

Problem Solving • Analytical Thinking • Technical Communication • Team Collaboration • Remote Collaboration • Continuous Learning • Adaptability • Attention to Detail • Time Management